MME 356 – MARINE ELECTRICAL TECHNOLOGY AND ELECTRICAL MEASUREMENTS (2, 0, 2)

Course objectives:

1. To provide insight into electrical propulsion system.
2. Give an understanding of how maintenance of electrical equipment is done.
3. Explain the basic concepts on measuring instruments.
4. Explain the classification of requirements for electrical equipment for steering gears, electrical control from wheelhouse for electro-hydraulic steering gear.

Content:

1. **Essential equipment and special circuits** - Classification of requirements for electrical equipment for steering gears, electrical control from wheelhouse for electro-hydraulic steering gear, all electrical steering gear circuit and control equipment, navigation lights. Miscellaneous marine equipment and alarm system-electrical engine room telegraph, remote indicating revolution counter, remote helm indicator, salinity indicators, mist detectors, carbon di-oxide recorder, electrical equipment for watertight door operation, centralized alarm system in machinery space, for the above, fire alarm system for engine room, holds and accommodation, high temperature alarms low- and high-level alarm.

2. **Electric propulsion** – systems, differences, and relative merits, alternating current, diesel electric and turbo electric propulsion system engines, excitation system and control.

3. **Maintenance of electrical equipment** - detection of faults and repair, preventive maintenance and periodic survey of equipment. Localization of cable faults. Classification of requirements for spares on board.

4. **Basic of measuring instruments** - deflection, control and damping devices moving coil, moving iron, dynamometer and thermocouple type of ammeter, voltmeter and wattmeter-their construction and other details, extension of scales of a meter. Single phase and three phase measurements by wattmeter. Measurement of energy. Measurement of speed, frequency and phase difference. Measurement of resistance, inductance and capacitance by Bridge Method. Magnetic measurement. Transducers and its application in the measurement of pressure, flow, temperature etc. Simple electronics measuring devices, such as VTVM, CRO, IC tester, signal generator.

Reading material:

Suresh Chandra Misra: Design Principles of Ships and Marine Structures, CRC Press, ISBN 9781482254464, 2015

CHAPTER 3 - MAINTENANCE OF ELECTRICAL EQUIPMENT

3.1 **MAINTENANCE**

Wear and tear in equipment is unavoidable. Equipment would usually have finite life span before eventually being retired from operations. The good news is that equipment do not just go end-of-life overnight. On the contrary, equipment are designed to be durable enough to last long periods. However, the life of equipment depends largely on how it is used and maintained. Within an equipment's life cycle, there is ample time to periodically check its condition and keep it at acceptable levels to maximize its use.

Maintenance is one thing that keeps any mechanical equipment or machinery going. Weather it is a small machine or a large structure, efficient maintenance can help with prolonged life and favourable outcome. On a ship, maintenance is one thing that keeps machinery up to date and in smooth running condition. In a ship's engine room, where the highest number of ships' machines are located, engineers and crew carry out maintenance for safe and efficient operation. Each machine on board a ship requires maintenance which must be carried out at regular intervals of time.

In earlier days, the number of crew members and engineers on a ship were large so maintenance was carried out fast and easily. However, at present, the number of crew members and engineers on board ships have reduced drastically. Many ships carry only 3-4 engineers on board. Additionally, the time required to carry out maintenance on ships has reduced. However, maintenance requires manpower and time which may not be available all the time as the number of crew members has reduced while the amount of machinery has increased. It is for this reason why it is important to plan the maintenance of ships' machinery so that machinery can always be kept in good working condition. Efficient planning and adequate usage of equipment are key to productive maintenance.

The following are the types of maintenance programmes carried out on ships:

(a) **Condition maintenance system:** In this system, machinery parts are checked regularly. With the help of sensors, etc., the condition of machinery is accessed regularly, and maintenance is done accordingly. This system requires experience and knowledge as wrong interpretation may damage the machinery and lead to costly repairs which may not be acceptable by the company.

(b) **Preventive maintenance:** Preventive maintenance is famously known as Planned Maintenance System or PMS. In this type of system, the maintenance is carried out as per running hours like 4000 hrs, 8000 hrs, etc., or by the calendar intervals like semi-annually, yearly etc. of the machinery. The maintenance is carried out irrespective of the condition of the machinery. Machine parts must be replaced if it is written in a schedule, even if they can still be used. A crew inspects a ship and carries out preventive measures according to a conditional analysis and maintenance schedule to keep equipment in good working condition. Examples of preventive maintenance include periodically cleaning and lubricating components, as well as projects related to surface preparation and coatings using marine temporary climate solutions.

Preventive maintenance can be sub-divided into planned maintenance and scheduled maintenance. Both planned and scheduled maintenance are preventive maintenance (PM) strategies that involve proactive activities. The two maintenance methodologies complement each other, with scheduled maintenance picking up where planned maintenance ends. However, despite their overarching similarities, they are not the same.

Planned maintenance refers to a maintenance process that details which resources will be needed to solve a particular equipment malfunction. The goal is to determine which tasks will be completed and how they will be completed. The process of planning maintenance involves identifying potential equipment issues, outlining steps required for upkeep, and procuring necessary tools and parts for execution. Equipment inspections, process descriptions, ordering replacement parts, and work prioritization make up the planning process of planned maintenance and are overseen by a maintenance planner.

On the other hand, scheduled maintenance refers to a maintenance process that details who will complete maintenance tasks and when. It focuses on when planned maintenance tasks will be completed and by whom. The goal is to ensure all maintenance activities are completed within a given timeframe. Work orders can be scheduled via paper, spreadsheets, or computerized maintenance management software (CMMS). Scheduled maintenance is completed at repeated intervals according to manufacturer's recommendations or per work requests.

Despite the differences in planned and scheduled maintenance, neither of the two maintenance strategies can work without the other. Planned maintenance without proper scheduling can result in unplanned downtimes when teams are not sure who is responsible or how work on certain tasks should be done. Some bodies recommend that at least 90 percent of maintenance activities be planned maintenance. Conversely, scheduled maintenance without proper planning results in shoddy workmanship due to a lack of the necessary materials, work tools, and documentation.

The benefits of planned maintenance and scheduled maintenance are:

- **Reduced downtimes:** When maintenance tasks are planned and scheduled, small problems can be taken care of before they develop into major issues or equipment failures. Planned and scheduled maintenance allows maintenance teams to gather data and prioritize work orders depending on their urgency.
- **Reduced maintenance costs:** For critical equipment, the more they deteriorate the more expensive it becomes to repair or replace them. Planning and scheduling maintenance tasks in the initial stages of a problem significantly lower the maintenance costs.
- **Improved equipment lifespan**: Keeping equipment in good working condition enables them to serve for long periods. Proper maintenance extends the lifespan of equipment.
- **Reduced workplace disasters:** When equipment are in good condition, workplace safety is improved, and workplace disasters are minimised.
- **Improved workplace culture:** Reduced downtimes means more engaged employees and positive workplace culture. It also makes collaboration among different teams much easier.
- **Maintenance is carried out at times most favourable to plants.**
- **Replacement can be forecasted and executed without disruption of activity.**
- **Specialist advice can be obtained at opportune times.**

(c) **Breakdown or corrective maintenance:** In this system, maintenance is carried out when machinery breaks down or fails. This is not a good method as situations may occur wherein the machinery is required in emergency. The only advantage of this system is that the working of machinery parts is used to its full life or until it breaks down. This system might get costly as during breakdown, several other parts may also get damaged.

(d) **Predictive maintenance:** This is a condition-based maintenance that involves the proactive, nonintrusive analysis of equipment or machinery using monitoring systems with statistical processes that determine when components will need corrective maintenance. Technologies utilized include infrared, corona detection, vibration analysis, acoustic, and oil analysis. Predictive maintenance provides the needed time to plan maintenance tasks efficiently while mitigating high costs associated with random failures.

The primary benefits of predictive maintenance for ships are summarised as follows:

(i) Ability to schedule corrective maintenance before vessels are on open waters, where it is more difficult and costly to conduct repairs. Because the technologies used in condition monitoring are like surveillance systems that warn of looming failures, they help workers predict the maintenance cycle of equipment or machinery.
(ii) It provides knowledge about parts to stock in advance
(iii) Reduced overhead cost
(iv) Energy savings
(v) Overall efficiency improvements
(vi) Reduced maintenance and repair costs
(vii) Improved productivity
(viii) Fewer component failures

While preventive maintenance is vital, and some accidents are unavoidable, predictive maintenance fills in the gaps that preventive maintenance leaves. Preventive maintenance only occurs in accordance with a program's schedule. Predictive maintenance technologies monitor in-service equipment, which workers cannot do throughout the day, and let crewmembers know when a component needs attention before it fails.

All maintenance works done on ships must conform to the International Safety Management Code or ISM Code for Ships. This code was adopted and integrated as part of the SOLAS (Safety of Life at Sea) convention in 1994.

In other words, it can be highlighted that the ISM Code is an intrinsic part of the International Maritime Organisation (IMO) in its efforts to ensure, maintain and effectuate safety for seafarers as well as simultaneously providing a pollution-free zone for the entire marine sector. It can be said that the code focuses on bringing marine safety to a common platform for vessels of all nationalities. This eliminates any discrepancies that could arise about the maintenance of a much-needed safety protocol. According to the stipulations of the code, it is mandatory that all ships follow this code. To execute its appropriate functionality, the International Safety Management Code is supported by a Safety Management System. This system, abbreviated as SMS, details the various requirements that need to be followed. Some of the requirements are:

- Establishment of a managerial committee to oversee the various proceedings.
- Ensure that managerial officers carry out their outlined duties appropriately.
- Corroborate the differences between outlined responsibilities and actual performance to resolve them.
- Audit vessels – both internally and externally to eliminate all potential safety problems.

The latter aspect comes under the ambit of the planned maintenance system. It is expected of every shipping corporation that it carries out auditory analyses of its safety management system and enforces the same, wherever lacking. While the internal auditing is carried out by the company itself, the external auditing is carried out, every two to three years, by officials of the country where the vessel is registered to. If the vessel has successfully incorporated all the safety requirements, then the officials issue it with a certification of safety management or the safety management certificate which brings the entire chain of a process to a fruitful completion.

When developing a maintenance procedure for a particular ship, the shipping company takes the following points into consideration:

- Maintenance guidelines and specifications given by the manufacturer.
- History of equipment including failures, defects, damagers, and remedial action.
- Guidelines mentioned in the ISM code.
- Age of the ship
- Third party inspections
- Consequences of failure of equipment on safe operation of the ship
- Critical equipment and systems
- Intervals of maintenance

3.2 **DETECTION OF FAULTS**

One of the most important qualities that a marine engineer must have is to know and understand his machinery very well. Before breaking down completely, each machinery will show a variety of signs and symptoms indicating the type and severity of the fault. Along with knowing the right procedures to operate ship's machinery, mariners must also know how to identify and troubleshoot any problem in the engine room.

The following are ways in which a mariner can identify and rectify faulty machinery:

(a) **Sound**

Sound is by far the most prominent factor which draws seafarer's attention towards a troubled part or machinery. If you are a good watchkeeper, it will be easy to figure out the difference between normal running sound and problematic sound even when you are not near the machinery. For e.g. A "hissing" sound will indicate leakage, a loud knocking sound will indicate loosen or broken parts, a high wobbling sound will indicate obstruction etc. Unfortunately, there is no guide to learn these sounds. Only through experience can one master such skills.

(b) **Abnormal smell**

Another powerful indication, which can be easily detected by human senses, is that of abnormal smell coming from machinery or systems. When you sense heavy/strong smell in a vicinity, it can

be due to leakage of oil, fire, effects of high temperature, etc. A burning smell near a motor is an indication of an increase in temperature of its coil. You can detect the smell of heavy oil even if you are not able to see it. Similarly, steam leakage will leave a dampen smell. It is only while working onboard ships, that one can know and understand different smells indicating a variety of problems with machinery.

(c) **High vibration**

All machinery systems with moving parts generate vibration. One of the most neglected maintenance jobs for machinery onboard ships is that of vibration analysis. Many shipping companies do not include it in its planned maintenance system. Even the timely checks for tightening the foundation bolts for any machinery are not included in some PMS. Every machinery will have its own frequency of vibrations. It is important to keep track of any increase in the vibration of machinery, which if ignored, can lead to severe damages in the long run. Any change is the vibration of machinery can easily be felt on board ships. This is a sign which should never be ignored.

(d) **Leakages**

Leakages are a result of faulty piping or machinery systems. They are easy to identify onboard ships. Never ignore leakage from any kind of machinery as it can lead to spills, fire, flooding, and other major accidents. If you find oil-water or air leak in machinery, do try to rectify it immediately or mark it as important to check during the next maintenance schedule depending on its severity.

(e) **Smoke**

Every machinery with a combustion chamber can be judged for its performance by checking the exhaust smoke for its colour and density. Exhaust smoke of main engine, auxiliary engine, boiler etc. must be monitored to know the condition of such equipment. Black smoke indicates a problem in the fuel injection system and improper combustion (lack of air etc.) whereas white smoke indicates water ingress in fuel.

(f) **Abnormal parameters**

Abnormal or fluctuating parameters are mainly related to machinery faults. It is important to keep a track of all machinery parameters onboard ships by comparing the readings in a logbook to the data of previous dates. While taking rounds, any deviation in the parameters must be taken seriously by taking proper investigation and preventive actions.

(g) **Alarms**

Every alarm indicates a problem, major or minor. Alarms are installed on equipment for the purpose of informing operators about faults. Never ignore an alarm related to any kind of machinery. An oil mist detector alarm in the main or auxiliary engine, even when other parameters are normal (crankcase temperature, scavenge temperature, etc.), must be taken seriously. Many incidents have been reported for crankcase explosion when OMD alarm has sounded but the crew ignored it because other parameters were seen as normal.

(h) **Observing problems in connected systems**

In a ship's engine room, most of the machinery or systems interlinked. Consequently, a problem in one equipment or system can affect others. Therefore, if a problem is observed in one system, other connected systems must be checked. For e.g., when there is a problem in an expansion tank, say, level suddenly going down, do check any leakage in the main engine, generator, or air compressor connected to it. A leakage in jacket water of the engine will lead to air going into the expansion tank with high pressure during the compression stroke and emptying the expansion tank from a vent or other openings.

(i) **Change in amperage**

More than 80% of the machinery on a cargo ship are electrically operated i.e., from ship's generated power. Ensure to check the current of all the electrically operated machinery and pumping systems. A high current for a purifier indicates a problem in the clutch drum or transmission gear. Similarly, high auxiliary blower current indicates scavenge pressure inside the engine is more than that supplied by the fan. Since in most ships, the auxiliary blower fans are operated manually, the fan must be switched off when the pressure is reached or when the current crosses the marked limit.

Knowing your designated machinery inside-out will help you identify the minutest change in its performance. Learning about its history, reading its maintenance reports, and keeping routine checks will give you an idea as to how your machinery acts and performs under different conditions. This would make it easy for you to recognize any fault in your machinery system when it operates differently from its usual working pattern.

3.3 **SPARES ONBOARD**

If a sudden operational failure of equipment or technical systems onboard may result in a hazardous situation, that equipment or system is considered critical. Master and chief engineer must draw up a list of critical ship functions and critical equipment (main and standby). These critical lists are to be formalised on board and sent to the appropriate office. The agreed list of critical equipment must be accurately reflected and identified in the vessel's PMS. This list will always include the supply of electrical power to essential services, propulsion, main engine, steering gear and generators. The list of critical ship functions may also include ship specific functions, for example the capability to recover from a semi-submerged condition for a heavy lift vessel.

(a) **Critical standby equipment**

Working from the basis of the critical machinery list, a list of critical stand-by arrangements should be drawn up. The critical stand-by arrangements list should be decided using the criteria of "Will single critical stand-by arrangement failure compromise one of the critical machinery functions". The critical stand-by arrangement list will generally include:

1. Electrical generating plant
2. Main air compressor sets
3. Control air compressor sets
4. Main and auxiliary engine circulating systems pump sets
5. Main engine electrical blower
6. Main switchboard and components
7. Main and auxiliary engine system filters
8. HFO & LO pumps
9. Feed water system pump sets

10. Boiler circulating pump sets
11. Boiler water gauge glass sets
12. Boiler safety valve sets
13. Steering gear pump sets
14. Rudder, shafting and propeller systems
15. Control engineering items
16. Ships side valves
17. Windlass and mooring sets
18. High and low level alarms
19. Stand-by whistle, siren horn etc
20. Critical idle functions (equipment which is safety critical but not in day-to-day use).
21. Quick closing valves, remote stops and switches
22. Over-speed trip arrangements
23. First start arrangement
24. Emergency electrical generating plan
25. Emergency lighting
26. Emergency compressor
27. Emergency fire pump
28. Fire fighting and detection systems
29. Safety and lifesaving equipment
30. B.A. Compressor
31. Emergency steering arrangement
32. Main & auxiliary engine shutdown arrangements.

It is the responsibility of the chief engineer to set the minimum level stock listing, customised for each vessel and to formally agree with the fleet superintendent on the content of the list. The chief engineer must then maintain the agreed minimum stock levels and to ensure, as appropriate, the necessary requisitions. Deficiencies are to be brought to the attention of the master and the technical department of the relevant management office as "critical spares inventory monthly report". This report must be completed and returned to the managing officer.

Critical spares onboard a ship can be used in two ways namely, proactively, and reactively. Proactively, spares are used onboard during planned maintenance to ensure the reliability of the safety critical equipment to mitigate the occurrence of hazardous situations. Reactively, spares are used onboard during unplanned maintenance to repair damaged safety critical equipment to prevent a hazardous situation from escalating to a more serious uncontrolled incident.

Refer to the "Guidance information on spear gear" on the number of spares required for various ship equipment. For effective management of spares, spares management systems (SMSs) are used.

3.4 **LOCALIZATION OF CABLE FAULTS**

Cable routes are influenced by a variety of ambient parameters. A cable route can consist of multiple diverse cable parts of diverse designs and types. Depending on the voltage level, the required load capacity, and available accessory and installation technology, cables with plastic insulation or mass-impregnated paper insulation are used.

Cable damage can be due to various causes. The most common sources of faults include end of service life and external influences as well as improper assembly. When these types of cable faults occur, the defective spot needs to be detected as quickly as possible and the fault rectified to minimise the duration of the failure. The following are the types of faults that can occur in cables.

(a) **Short-circuit:** Damaged insulation leads to a low-resistance connection of two or more conductors at the fault location.

(b) **Earth fault / short-circuit to earth:** Earth faults or short-circuits to earth are low-resistive connections to the earth potential. The double earth fault is another type of fault; this fault shows two earth faults on different phases with separated bases.
(c) **Cable sheath faults:** Damage to the outer cable sheath does not always lead directly to faults. However, it can cause long-term cable faults, among other things, because of moisture penetration and insulation damage.
(d) **Cable breaks:** Mechanical damage and ground movements can lead to breakage of individual or multiple conductors.

3.4.1 Process steps and methods for cable fault location

Fault location is carried out methodically following a logical procedure, in four steps.

(a) **Fault analysis:** Fault analysis makes it possible to determine the characteristics of the fault and the further steps to take.
(b) **Pre-location:** During pre-location, the fault position is determined as precisely as possible.
(c) **Tracing and pin-pointing:** The objective of the subsequent pin-pointing process is to determine the precise fault location in order to limit excavation work and thus minimise costs and the repair time.
(d) **Cable identification:** Next comes cable identification, as it is necessary to identify the correct cable among the multiple cables at the fault location. This is especially important if the fault is not visible from outside.

The objective of fault location is to locate a cable fault as quickly and precisely as possible to provide the ideal basis for fast repair and reconnection. To this end, manufactures have produced various equipment for rapid localization of cable faults.